ZARAILINK
TRADE INTELLIGENCE

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Introduction

Pakistan's traders rely on scattered tools and disconnected data to find global buyers and suppliers. ZaraiLink fixes this -one AI-powered platform for partner discovery, trade intelligence, and verified contacts.

Objectives

- Unified AI search (natural language, products, HS codes)
- NLU pipeline (intent, product, price/quantity parsing)
- Supplier & buyer intelligence (pricing + shipment analytics)
- Compare up to 4 entities with PDF export
- Verified Pakistani trade directory with contacts
- Token-based paywall for monetization

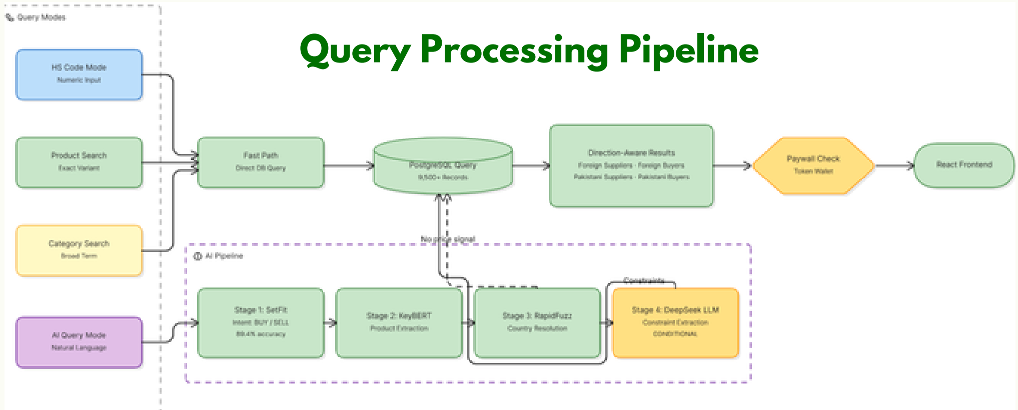
Tech Stack

Python · React.js · Redis
Django · PostgreSQL · Docker

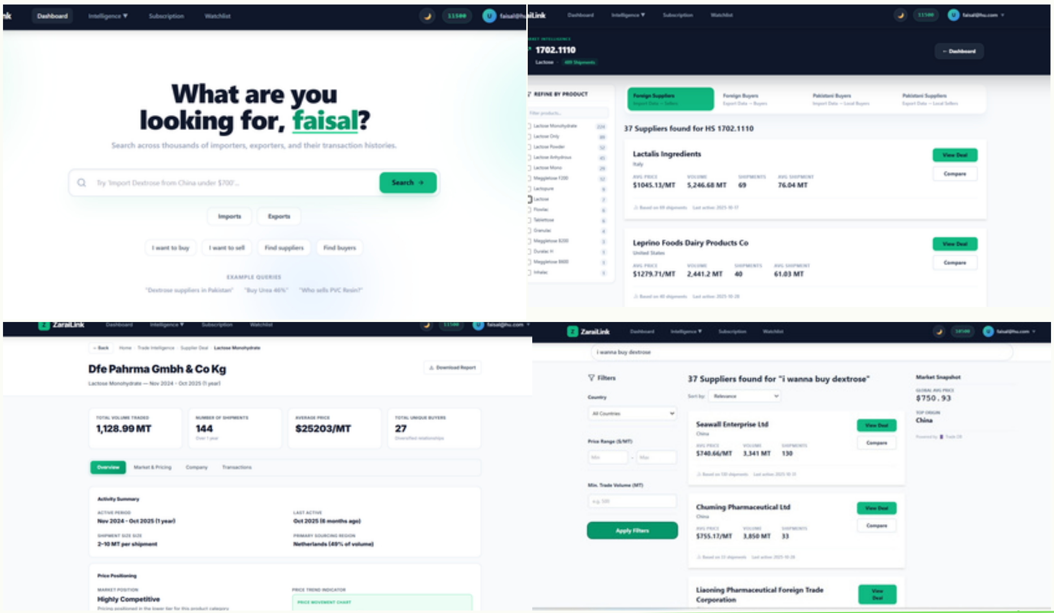
Methodology

ZaraiLink uses a hybrid search system where direct product/HS code queries take a fast, non-AI path for instant results.
Natural language queries pass through a 4-stage NLU pipeline to extract intent, product, location, and pricing before querying the database.

Query Processing Pipeline



Snapshots from application



Key Features

Results and Metrics

Platform Scale

89.4% Intent Detection
9500+ Trade Records
354 Training Queries
4 NLP Models

Search Performance

Fast Path is 22× Faster

Fast Path — 40ms
NLU Cached — 200ms
NLU cold — 900ms

Future Work

- Mobile app for field-level traders and SMEs
- Urdu language support for local accessibility
- Real-time price alerts and market movement notifications
- Predictive analytics for demand forecasting and price trends

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Conclusion

ZaraiLink transforms Pakistan's fragmented trade discovery into an intelligent, data-driven platform. Beyond search, it delivers deep Deal Intelligence covering procurement behavior, pricing, benchmarking, partner networks, and transaction history for confident decision-making. Its hybrid search using a Fast Path and a 4-stage NLU pipeline delivers insights from 9,500+ real shipment records in milliseconds, enabling faster partner discovery and smarter global trade decisions.